Problem 5.15

A 3% discount is offered for cash payment of a \$2500 bill, due at the end of 90 days. At what rate of simple interest earned over the 90 days is cash payment made? Use the Banker's Rule.

The discount is

$$0.03(2500) = 75.$$

So the cash payment is

$$2500 - 75 = 2425.$$

This means that paying early earns \$75 of interest on an investment of \$2425 for 90 days.

We use the simple interest formula:

$$I = Prt,$$

where $I = 75$, $P = 2425$, and, by Banker's Rule,

$$t = \frac{90}{360}.$$

Substituting into the formula:

$$75 = 2425 \cdot r \cdot \frac{90}{360}.$$

Solving for r,

$$r = \frac{75}{2425\left(\frac{90}{360}\right)}.$$

Therefore,

$$r \approx 0.1237.$$

So, the simple interest rate is

$$\boxed{12.37\%}.$$